Ziwei Gu

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Overview

Fifth-year Computer Science Ph.D. student at Harvard working on **LLM evaluation** and **human-AI interaction**. I design benchmarks and multi-turn evaluations that test how LLMs handle realistic user requests, such as ambiguous or mixed-language queries. I also build AI-powered interfaces that help people process information at scale while noticing and recovering from AI errors and steering the models.

Education

Harvard University — *Cambridge, MA*
Ph.D. Computer Science — *Aug 2022 – May 2027 (Expected)*
Advisor: Elena Glassman

Cornell University — *Ithaca, NY*
B.A. Computer Science, Magna cum laude — *Aug 2017 – May 2021*
B.A. Mathematics — *Aug 2017 – May 2021*

Publications

** Equal contribution*

Conference Papers

P.6 **Ziwei Gu**, Joyce Zhou, Ning-Er (Nina) Lei, Jonathan K. Kummerfeld, Mahmood Jasim, Narges Mahyar, and Elena L. Glassman. AbstractExplorer: Leveraging Structure-Mapping Theory to Enhance Comparative Close Reading at Scale. *In ACM Symposium on User Interface Software and Technology (UIST)*, 2025.

P.5 **Ziwei Gu**, Ian Arawjo, Kenneth Li, Jonathan K. Kummerfeld, and Elena L. Glassman. An AI-Resilient Text Rendering Technique for Reading and Skimming Documents. *In ACM CHI Conference on Human Factors in Computing Systems (CHI)*, 2024. **Patent pending**

P.4 Katy Ilonka Gero, Chelse Swoopes, **Ziwei Gu**, Jonathan K. Kummerfeld, and Elena L. Glassman. Supporting Sensemaking of Large Language Model Outputs at Scale. *In ACM CHI Conference on Human Factors in Computing Systems (CHI)*, 2024. **Honorable Mention Award (top 5%)**

P.3 Jing Nathan Yan, **Ziwei Gu**, and Jeffrey M. Rzeszotarski. Tessera: Discretizing Data Analysis Workflows on a Task Level. *In ACM CHI Conference on Human Factors in Computing Systems (CHI)*, 2021.

P.2 **Ziwei Gu***, Jing Nathan Yan*, and Jeffrey M. Rzeszotarski. Understanding User Sensemaking in Machine Learning Fairness Assessment Systems. *In WWW'21: The Web Conference (WWW)*, 2021.

P.1 Jing Nathan Yan, **Ziwei Gu**, Hubert Lin, and Jeffrey M. Rzeszotarski. Silva: Interactively Assessing Machine Learning Fairness Using Causality. *In ACM CHI Conference on Human Factors in Computing Systems (CHI)*, 2020.

Workshop Papers

W.2 Ajit Mallavarapu and **Ziwei Gu**. Sampling Reveals Style: Unsupervised, Training-Free Discovery of Prompt-Conditional Stylistic Axes in LLM Activations. *In the 9th Workshop on Analyzing and Interpreting Neural Networks for NLP at EMNLP (BlackboxNLP)*, 2026.

W.1 **Ziwei Gu***, Gauri Jain*, Hongwen Song*, Isak Diaz*, Margaux Masson-Forsythe*, and Jorge Valdes. BiomeAzuero2022: A Fine-Grained Dataset and Baselines for Tree Species Classification with Ground Images. *In the 37th AAAI Conference on Artificial Intelligence (AAAI-23) AI for Social Good Workshop*, 2023.

Extended Abstracts

EA.1 **Ziwei Gu**, Owen Raymond, Naser Al Madi, and Elena L. Glassman. Why Do Skimmers Perform Better with Grammar-Preserving Text Saliency Modulation (GP-TSM)? Evidence from an Eye Tracking Study. *In ACM CHI Conference on Human Factors in Computing Systems Late Breaking Work (CHI LBW)*, 2024.

Preprints

PP.1 **Ziwei Gu***, Michael S. Lee*, Drew Rein, Yash Maurya, Vikneswaran Kumaran, Darshini Ramiah, Chenguang Wang, and Udari Madhushani Sehwag. SG-LegalClarify: a Large-scale Benchmark for Legal Reasoning and Ambiguity Recognition in Singapore Law. *Under review at ICLR 2027.*

Research Experience

Harvard University – Graduate Researcher *Cambridge, MA*
Advisor: Elena Glassman *Aug 2022 – Present*

- Invented GP-TSM, an LLM-powered recursive sentence compression method that identifies successive levels of less important detail and renders them in lighter gray for grammar-preserving skimming; readers (N=18) scored higher on GRE reading comprehension in 12% less time than with plain text [P.5, EA.1].
- Developed LLM-powered interfaces, grounded in Structure-Mapping Theory, for comparative reading of 1,000+ research abstracts [P.6] and sensemaking of LLM outputs at scale [P.4].
- Proposed and validated an unsupervised, training-free method for discovering prompt-conditional stylistic axes in LLM activations; top axes matched human-requested styles with up to 72.8% precision [W.2].

Cornell University – Undergraduate Researcher *Ithaca, NY*
Advisor: Jeff Rzeszotarski *2019 – 2021*

- Developed Silva, an interactive system that uses causal graphs to help users reason about sources of unfairness in machine learning datasets and models [P.1].
- Led a mixed-methods study (logs, think-alouds, interviews) of how people make sense of machine learning fairness issues with exploratory vs. recommendation-based tools [P.2].

Industry Experience

Scale AI – Human Frontier Collective Fellow *Remote*
Mentors: Michael Lee, Udari Madhushani Sehwag *Dec 2025 – Present*

- Co-led the development of SG-LegalClarify, a benchmark of 1,100 lawyer-authored questions across 10 legal domains testing legal reasoning, ambiguity recognition, and selective clarification [PP.1].
- Built its multi-turn evaluation pipeline with a tool-gated user simulator and LLM-as-judge grading; found substantial headroom across 19 frontier LLMs (best: 69.8% of mandatory rubric items, 77.8% with web search).
- Drove cross-functional coordination with multiple stakeholders to organize 20+ contributors across data collection and meta-evaluation.

Adobe Research – Research Scientist Intern *San Jose, CA*
Mentors: Doga Dogan, Gromit Yeuk-Yin Chan, Eunyee Koh *Sep 2026 – Present*

- Experimented with LLM simulation strategies to estimate the visibility of unpublished drafts in AI search.
- Developed a Generative Engine Optimization (GEO) Google Docs extension that lets content creators understand and test how their edits affect AI visibility as they write.

Autodesk Research – Research Scientist Intern *Toronto, ON, Canada*
Mentors: John Thompson, Justin Matejka *May 2026 – Sep 2026*

- Proposed Ink-to-3D-Control, a novel interaction technique that uses a VLM to turn designers' ink and speech into manipulable 3D controls that expose an agent's interpretation before any geometry changes.
- Built and evaluated InKontrol, a full-stack generative 3D-editing system (React Three Fiber, Python, CadQuery) that surfaces clarification questions and an editable build plan before generating 3D geometry; in a user study (N=12), it cut median regeneration cycles per task from 4.0 to 1.2.

Lyft – Data Scientist Intern *San Francisco, CA*
Mentors: Mengying Yang, Bomee Park *2020*

- Clustered millions of rider transactions into user profiles to size the Lyft Family opportunity; co-proposed and advocated for the feature with data, and it remains live in the app today.
- Modeled driver churn with a Markov chain; quantified incentive effects to inform engagement strategies.

Honors & Awards

Honorable Mention Award (top 5%), ACM CHI [P.4] *2024*
Magna cum laude, B.A. Computer Science, Cornell University *2021*
Statistics Faculty Award, Cornell University *2018*

TEACHING EXPERIENCE	*Teaching Fellow*, Conceptualizing, Building, and Evaluating Usable Novel Interactive Systems, Harvard CS	*2026*
	Head Teaching Fellow, Engineering Usable Interactive Systems, Harvard CS	*2023*
	Graduate Teaching Assistant, Operating Systems, Cornell CS	*2021*
	Teaching Assistant, Machine Learning, Cornell CS	*2019 – 2020*
	Teaching Assistant, Computer System Organization and Programming, Cornell CS	*2019*
	Teaching Assistant, Object-Oriented Programming and Data Structures, Cornell CS	*2018 – 2019*
MENTORING	Ajit Mallavarapu (University of Washington) [W.2], Samhitha Goutham (Anna University), Kanishk Tantia (Harvard University), Azwad Alam (University of Southern California)	
ACADEMIC SERVICES	*Reviewer* CHI (2024, 2025, 2026, 2027), UIST (2024, 2025, 2026), BlackboxNLP Workshop at EMNLP 2026, VIS 2025, Creativity & Cognition 2025	
	Student Volunteer IUI 2023	
SKILLS	**Eval Engineering:** LLM-as-judge, benchmark/rubric design, meta-evaluation, user simulation	
	ML & Data: Python, PyTorch, agentic workflows, experimental design, tool calling & structured outputs, RAG	
OUTREACH ACTIVITY	*Fellow (Intellectual and Cultural)*, The Student Center at Harvard Griffin GSAS	*2023 – 2025*
	Resident Advisor, Clara Dickson Hall, Cornell University	*2019 – 2021*
	Project Lead, Cornell Data Science Project Team	*2018 – 2020*

Latest update: 11/2026